Results draft: Fernhill Software, version 1 of 3 (the enthusiast's model)

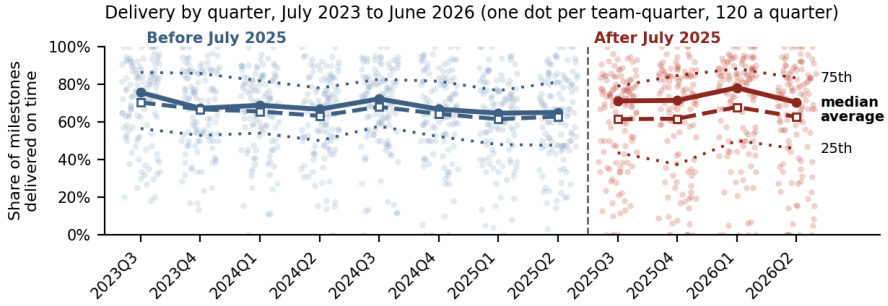
1. The question and the prediction

My question is causal. Would replacing engineer-managers with professional managers, technical leads and AI raise delivery (the share of a team's committed milestones delivered by their dates)? A before-and-after can answer only a descriptive question, so that is this draft's question. How did delivery and people-skill ratings (the share of favorable survey answers about a manager) move from the eight quarters before July 2025 to the four after? My model predicts delivery up about 5 points after a first-quarter dip, ratings up, and the rise holding on hard quarters. The Brief didn't say which summary of delivery the 5 points meant, so I read it as the average.

2. The charts

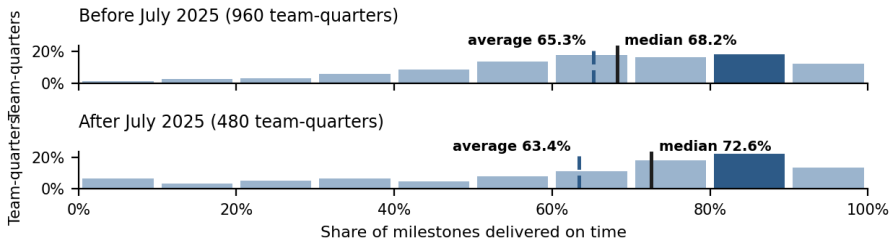
The first chart plots every team in every quarter as a faint dot and draws the median, the average and the 25th and 75th percentiles through them, with a dashed line where the reorganization began.

> **This chart answers:** how did delivery move, quarter by quarter, before and after the reorganization?
>
> Delivery is a measurement (a share), and the quarter is a date. Each dot is one team in one quarter, 120 a quarter. The solid line joins the quarterly medians and the dashed line joins the quarterly averages. The dotted lines join the 25th and 75th percentiles, with a quarter of teams below the first and a quarter above the second.



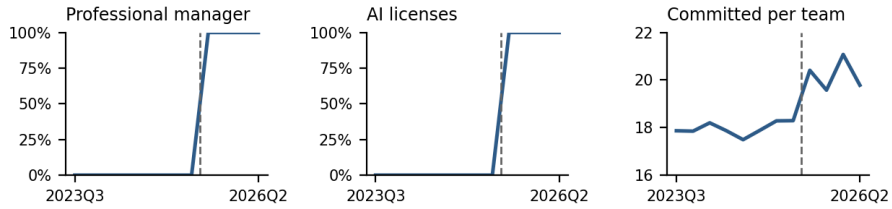
The quarters are noisy, so the second chart pools them into before and after, stacks the two on one shared axis, and marks the average and the median on each.

> **This chart answers:** how is delivery spread across team-quarters before and after July 2025, and where do the average, the median and the most common band fall?
>
> Delivery is a measurement (a share). Each bar is the share of that period's team-quarters in a ten-point band (960 before, 480 after). The dashed line marks the average, the solid line marks the median, and the darkest bar marks the most common band.



Delivery was not the only record that moved in July 2025, and the third chart shows what else did.

> **This chart answers:** what else changed in July 2025?
>
> Each point on a line is one quarter, averaged over the 120 teams. The first two panels plot the share of teams with a professional manager (from a label) and with AI licenses (from a yes/no), and the third plots milestones committed per team (from a count).



Each panel is one record, averaged over 120 teams each quarter. The dashed line marks July 2025

3. The association

On average, delivery is lower after July 2025, at 63.4% against 65.3% before, a difference of 1.9 points. This is measured at the team-quarter level, over 1,440 team-quarters (960 before, 480 after). **But the middle moved the other way.** The median team-quarter delivered 72.6% after against 68.2% before, which is 4.4 points higher. The most common band (80% to 90%, by a hair before) did not move at all.

The three measures disagree because they answer different questions, and they agree only when a distribution is symmetric. The average is the balance point, the median is the middle, and the most common band is the peak. This distribution is skewed left, with a long tail of weak quarters, and the tail got longer. The weakest tenth of team-quarters delivered about 35% or less before and about 19% or less after, so the average follows the tail while the median follows the middle.

At the average, the fall is weak. Every after-quarter (61.4% to 67.8%) sits inside the range of the before-quarters (61.4% to 70.3%), which moved about 3 points from one quarter to the next. At the team level, the fall holds, barely. Of 120 teams, 68 delivered less after, and the median team fell 2.7 points. This is the middle of the team changes, not the change in the middle. Without the first after-quarter, the fall is 1.2 points, and weighted by milestones it is 1.3 points. At the team level, however, 56 of 120 teams delivered less without that quarter, and the median team's change is +1.0 point, so "lower" leans on one quarter.

Split by the quarter-close rating, team-quarters rated easy rose from 74.9% to 79.4% (541 before, 280 after), and those rated hard fell from 52.8% to 41.0% (419 before, 200 after). People-skill ratings rose from 0.69 to 0.79 (1,338 team-quarter ratings, with 102 blank and left out).

And the two changes are one variable. In all 1,440 team-quarters a professional manager and AI licenses go together, 480 with both and 960 with neither. The third chart's first two panels are the same line.

4. Against the model

I can't tell yet on delivery, because the summaries disagree. At the average, which is how I read the prediction, delivery is inconsistent with it. It is lower, not up 5 points, though the first-quarter dip did appear (61.4% against 62.8% the quarter before). At the median it is consistent (4.4 points up, near the 5), but I chose that reading after the average failed, so I don't count it as a pass, and AI alone would predict the same rise. Ratings are consistent (0.69 to 0.79, and 0.78 to 0.80 in each of the four quarters after, against 0.67 to 0.70 before). A honeymoon for new managers would predict the same rise but should fade, and this one held. Hard quarters are inconsistent, since the model said the rise would hold and they fell from

52.8% to 41.0%, but difficulty was rated after the outcome, so I can't tell yet whether that is a failure or an artifact.

But I can't tell whether the model failed or the comparison did. With the reorganization and AI arriving together, any difference belongs to both, and nothing in this file can split it.

5. What this does not establish

The rival I can test with this file is a change at the same moment, the year's customer migration, which every team took on from July 2025, because the file records how much work was committed. If it lowered delivery, I would expect more work committed after July 2025 in most teams, and a smaller share delivered even where more was delivered. Committed milestones rose from 18.0 to 20.2 per team-quarter (12%) and were higher in 94 of 120 teams, and delivered milestones rose from 11.6 to 12.8 while the share fell. That fits the migration, though it can't rule out AI leading teams to take on more.

The hard-quarter split is partly circular. Difficulty is rated at close, after the outcome, and quarters that delivered under 60% were rated hard 83% of the time, against 21% of the rest.

My Data & Methods concern, the ratings, changes little here, since blanks are 7% before and after, though a manager rated after one quarter may enjoy a honeymoon. Noticed, not pursued. Escalations fell from 9.9 to 7.5 a team-quarter, and lead departures rose from 3.6% to 11.9% a quarter. Departures may be a cause of weak quarters or a result of the reorganization, and this file can't tell which.

The decision needs a model that explains. Fernhill must choose what to keep, and a forecast for the package can't say which part delivered what, or why the weakest quarters got weaker while the middle improved.

What would separate the changes, using records Fernhill already keeps:

- Give the changes to different teams by lot, since AI licenses arrive in waves anyway. Some teams keep an engineer-manager without AI for a year, some get AI only, and some get the reorganization with AI.
- Set each group against its own eight quarters before, which the planning tool already holds, so the migration falls on every group alike.
- Rate difficulty at the quarterly planning review, before the work, instead of at close.
- Follow escalations and waits in the issue tracker, where the technical questions the model leaves to AI and the lead actually go.

Note on AI use

I used Gemini (Willamette account) for the averages and medians, the team-level changes, the crosstab, and the charts. It called the before-and-after difference "the effect of the reorganization." I changed that to "lower after July 2025," since nothing here separates the reorganization from AI or the migration. It gave the average alone. I added the median and the most common band, which tell a different story.

A note on length

This draft runs over the limit on purpose. It is longer than two pages and has three charts, where the rubric asks for one to two pages and one or two charts. The second chart is there to show that the average, the median and the most common band tell different stories, which is the point of section 3, and I would rather make that point than keep to the limit. Under the rubric that costs the draft something, and I accept the cost.